

The special unitary group, monotone metrics and gradient vector fields

An intertwined tale

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Space of (mixed) Quantum States

Start from ($n \times n$ complex) matrices.

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \in \mathcal{M}_n(\mathbb{C}) \quad (1)$$

On this algebra an involution $\dagger$ can be defined:

$$\dagger : \mathcal{M}_n(\mathbb{C}) \ni A \mapsto A^\dagger = \begin{pmatrix} \overline{a_{11}} & \dots & \overline{a_{n1}} \\ \vdots & \ddots & \vdots \\ \overline{a_{1n}} & \dots & \overline{a_{nn}} \end{pmatrix} \in \mathcal{M}_n(\mathbb{C}) \quad (2)$$

$\dagger$ has two eigenspaces:

$$\mathcal{V} = \left\{ \mathbf{a} \in \mathcal{M}_n(\mathbb{C}) : \mathbf{a} = \mathbf{a}^\dagger \right\}; \quad i\mathcal{V} = \left\{ \mathbf{b} \in \mathcal{M}_n(\mathbb{C}) : \mathbf{b} = -\mathbf{b}^\dagger \right\} \quad (3)$$

A **quantum state** ρ is a matrix in $\mathcal{V}$ whose eigenvalues are non-negative and sum to 1:

$$\overline{\mathcal{S}_n} = \{\rho \in \mathcal{V} : \rho \geq 0 \quad \text{and} \quad \text{Tr}(\rho) = 1\} \quad (4)$$

$\overline{\mathcal{S}_n}$ is the space of quantum states.

Quantum states “contain” probability vectors

$$\mathbf{p} = (p_1, \dots, p_n) \mapsto \rho_{\mathbf{p}} = \begin{pmatrix} p_1 & 0 & \cdots & 0 \\ 0 & p_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & p_n \end{pmatrix} \quad (5)$$

Stratification of the Space of Quantum States

$\overline{\mathcal{S}}_n$ is **not** a manifold, the following results hold:

Theorem

The spaces $\mathcal{S}_n^k$ of states of rank k , with $1 \leq k \leq n$ are smooth and connected manifolds in $\mathcal{V}$ of real dimension $2nk - k^2 - 1$.

Corollary

The boundary $\partial\mathcal{S}_n$ is the union of all strata with $k < n$. It is not a smooth manifold unless $n = 2$.

Moreover $\mathcal{S}_n \equiv \mathcal{S}_n^n$ is an open in Π_1

The Qubit

The qubit is the "simplest" quantum system.

$$\rho = \frac{1}{2} \left(\mathbb{I}_2 + \sum_{j=1}^3 x^j \sigma_j \right) \quad (6)$$

Where σ_j is the j -th Pauli matrix:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (7)$$

We have that:

$$\rho = \frac{1}{2} \begin{pmatrix} 1 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & 1 - x^3 \end{pmatrix}; \quad \det(\rho) = \frac{1}{4} (1 - |\mathbf{x}|^2) \quad (8)$$

The Qubit

Of course the determinant of ρ must be non-negative:

$$\det(\rho) \geq 0 \implies |\mathbf{x}|^2 \leq 1 \quad (9)$$

We have rank one (pure) states exactly when the equality is satisfied.

Remark

$\mathcal{S}_2$ can be seen as an open ball of radius 1 in $\mathbb{R}^3$, pure states lie on the boundary of this ball, thus $\overline{\mathcal{S}_2}$ can be seen as a closed ball of radius 1 in $\mathbb{R}^3$.

Group Actions

Let us populate the space of quantum states.

On $\overline{\mathcal{S}}_n$ we define an action of $\mathrm{SL}(n, \mathbb{C})$:

$$\alpha : \overline{\mathcal{S}}_n \times \mathrm{SL}(n, \mathbb{C}) \ni (\rho, g) \mapsto \alpha_g(\rho) = \frac{g\rho g^\dagger}{\mathrm{Tr}(g\rho g^\dagger)} \quad (10)$$

The orbits of α are the **strata** $\mathcal{S}_n^k$.

The special unitary group is:

$$\mathrm{SU}(n) = \left\{ U \in \mathrm{SL}(n, \mathbb{C}) : UU^\dagger = \mathbb{I} \right\} \quad (11)$$

If we "restrict" the action α to unitary elements of $\mathrm{SL}(n, \mathbb{C})$ we get:

$$\tilde{\alpha} : \overline{\mathcal{S}}_n \times \mathrm{SU}(n) \ni (\rho, U) \mapsto \tilde{\alpha}_U(\rho) = U\rho U^\dagger \quad (12)$$

The orbits of $\tilde{\alpha}$ are the so-called *manifold of isospectral states*.

Group Actions: infinitesimal version (I)

The Lie algebra of $SL(n, \mathbb{C})$ is $\mathcal{M}_n^0(\mathbb{C})$:

$$\mathcal{M}_n^0(\mathbb{C}) = \{A \in \mathcal{M}_n(\mathbb{C}) : \text{Tr}(A) = 0\} \quad (13)$$

Take the curve $g_t = e^{\frac{1}{2}(\mathbf{a} - i\mathbf{b})t}$ and compute:

$$\begin{aligned} \Gamma_{\mathbf{ab}}(\rho) &= \left. \frac{d}{dt} \frac{g_t \rho g_t^\dagger}{\text{Tr}(g_t \rho g_t^\dagger)} \right|_{t=0} = \frac{1}{2i} (\rho \mathbf{b} - \mathbf{b} \rho) + \frac{1}{2} (\rho \mathbf{a} + \mathbf{a} \rho) - \text{Tr}(\mathbf{a} \rho) \rho \\ &= [[\rho, \mathbf{b}]] + \{\rho, \mathbf{a}\} - \text{Tr}(\mathbf{a} \rho) \rho \quad (14) \end{aligned}$$

We have the decomposition

$$\Gamma_{\mathbf{ab}} = \mathbb{Y}_{\mathbf{a}} + \mathbb{X}_{\mathbf{b}} \quad (15)$$

$$\mathbb{Y}_{\mathbf{a}} := \Gamma_{\mathbf{a}0}, \quad \mathbb{Y}_{\mathbf{a}}(\rho) = \{\rho, \mathbf{a}\} - \text{Tr}(\mathbf{a} \rho) \rho \quad (16)$$

$$\mathbb{X}_{\mathbf{b}} := \Gamma_{0\mathbf{b}}, \quad \mathbb{X}_{\mathbf{b}}(\rho) = [[\rho, \mathbf{b}]] \quad (17)$$

Group Actions: infinitesimal version (II)

The Lie algebra of $SU(n)$ may be identified with:

$$\mathfrak{su}(n) \simeq \{\mathbf{a} \in \mathcal{V} : \text{Tr}(\mathbf{a}) = 0\} \quad (18)$$

$$[[\mathbf{a}, \mathbf{b}]] = \frac{1}{2i} (\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a}) \quad (19)$$

Take the curve $U_t = e^{-\frac{i}{2}\mathbf{b}t}$ and compute:

$$\tilde{\Gamma}_{\mathbf{b}}(\rho) = \frac{d}{dt} \left(U_t \rho U_t^\dagger \right)_{t=0} = \frac{1}{2i} (\rho \mathbf{b} - \mathbf{b} \rho) = [[\rho, \mathbf{b}]] \quad (20)$$

We get:

$$\tilde{\Gamma}_{\mathbf{b}} = \Gamma_{0\mathbf{b}} = \mathbb{X}_{\mathbf{b}} \quad (21)$$

$$\mathbb{X}_{\mathbf{b}}(\rho) = [[\rho, \mathbf{b}]] \quad (22)$$

In Classical Information Geometry we use Markov maps.

Theorem (Chentsov)

There exists a unique Riemannian metric tensor monotone under the action of Markov maps, and this is the Fisher-Rao metric tensor.

In Quantum Information Geometry we use completely-positive trace-preserving (**CPTP**) maps¹.

There is no analog of Chentsov theorem in the quantum case.

In the quantum setting we lose uniqueness.

¹A linear map $\phi : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_m(\mathbb{C})$ is CP if $\phi \otimes id_k$ from $\mathcal{M}_n(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ to $\mathcal{M}_m(\mathbb{C}) \otimes \mathcal{M}_k(\mathbb{C})$ is positive for every k .

Petz characterization of information metrics

A general characterization for information metric has been given by Petz:

$$G_{\rho}^f(A, B) = \text{Tr}(\mathbf{K}_{\rho, f}^{-1}(A)B) \quad A, B \in \mathcal{T}_{\rho}\mathcal{S}_n \quad (23)$$

Where:

$$\mathbf{K}_{\rho, f} = f(\mathbf{L}_{\rho}\mathbf{R}_{\rho}^{-1})\mathbf{R}_{\rho} \quad (24)$$

With:

$$\mathbf{L}_{\rho}(A) = \rho A \quad \mathbf{R}_{\rho}(A) = A\rho \quad (25)$$

And with f an operator monotone function² satisfying $tf(t^{-1}) = f(t)$ and $f(1) = 1$.

² f is monotone w.r.t. the partial order on $\mathcal{M}_n(\mathbb{C})$ given by: $A > B$ if $A - B$ is positive definite

Information metrics and the unitary group

Invariance under CPTP maps of information metrics implies their unitary invariance.

This in turn implies that the fundamental vector fields of the action $\tilde{\alpha}$

$$\mathbb{X}_{\mathbf{b}} = [[\rho, \mathbf{b}]] \quad (26)$$

are Killing vector fields for **all** information metrics.

The special unitary group is a sort of "universal symmetry" in Quantum Information Geometry.

The qubit again

Let us now turn to the qubit again. Recall that:

$$\det(\rho) = \frac{1}{4}(1 - |\mathbf{x}|^2) \quad (27)$$

Notice here that the determinant fixes the distance of a state from the center of the ball, and viceversa. Also the determinant of a state fixes his spectrum.

Remark

Isospectral surfaces are spheres of radius $0 \leq r \leq 1$, the fundamental vector fields of the action $\tilde{\alpha}$ will be tangent to these surfaces.

Regarding the action α (of $\mathrm{SL}(n, \mathbb{C})$) we have just two orbits, one being the interior of the ball and the other being the boundary.

The qubit again

For the qubit, the explicit form, in spherical coordinates, of a generic monotone quantum metric is

$$G_f = \frac{1}{1-r^2} dr \otimes dr + \frac{r^2}{1+r} \left(f\left(\frac{1-r}{1+r}\right) \right)^{-1} (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (28)$$

Where f is an operator monotone function $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ satisfying $f(1) = 1$ and $f(t) = tf(t^{-1})$.

An observation: BH metric

The so-called Bures-Helstrom metric G_{BH} is G_f with:

$$f(t) = \frac{1+t}{2} \quad (29)$$

Given $\mathbf{a} \in \mathcal{V}$ we can define the *expectation value function*:

$$e_{\mathbf{a}}(\rho) = \text{Tr}(\mathbf{a}\rho) \quad \mathbf{a} \in \mathcal{V} \quad (30)$$

It is possible to prove that:

$$G_{BH}(X, \mathbb{Y}_{\mathbf{a}}) = \mathcal{L}_X e_{\mathbf{a}} \quad \text{for every vector field } X \quad (31)$$

The $\mathbb{Y}_{\mathbf{a}}$ are gradient vector fields of expectation value functions w.r.t. G_{BH} .

An observation: BH metric

We have that:

$$\begin{array}{ccc} \text{Fund. v. fields of } \alpha & \longrightarrow & \Gamma_{\mathbf{ab}} = \mathbb{Y}_{\mathbf{a}} + \mathbb{X}_{\mathbf{b}} \longleftarrow \text{Fund. v. fields of } \tilde{\alpha} \\ & \uparrow & \\ & \text{Gr. v. fields of e. v. functions w.r.t. BH metric} & \end{array}$$

Remark

The Lie algebra of fundamental vector fields of α can be represented by the Hamiltonian vector fields together with the gradient vector fields of expectation value functions (w.r.t. Bures-Helstrom metric), equipped with the standard Lie bracket.

A general question

How general is this instance?

Specifically:

- Given a monotone metric tensor G_f on the space of quantum states, can the gradient vector fields of expectation value function w.r.t. G_f , together with $\mathbb{X}_b$ give rise to a representation of a Lie algebra?
- If so, does this algebra integrate to some group action on the space of quantum states?

How does this algebra look like?

There are some general features this algebra $\mathcal{A}$ must fulfill:

- It must be isomorphic, as a vector space, to $\mathcal{V}_0 \oplus \mathcal{V}_0$.
Let $\mathbb{Y}_{\mathbf{b}}^f$ be the gradient vector field of $e_{\mathbf{b}}$ w.r.t. G^f :

$$\Gamma_{\mathbf{ab}} = \mathbb{X}_{\mathbf{a}} + \mathbb{Y}_{\mathbf{b}}^f \quad (32)$$

- Given that $[\mathbb{X}_{\mathbf{a}}, \mathbb{X}_{\mathbf{b}}] = \mathbb{X}_{[[\mathbf{b}, \mathbf{a}]]}$ then $\mathcal{V}_0 \oplus \{0\}$ must be a Lie subalgebra of $\mathcal{A}$ isomorphic to $\mathfrak{su}(n)$

What about the commutator $[\mathbb{X}_{\mathbf{a}}, \mathbb{Y}_{\mathbf{b}}^f]$?

How does this algebra look like?

Consider:

$$\begin{aligned}\mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c &= \mathcal{L}_{\mathbb{X}_a} \left(\mathcal{L}_{\mathbb{Y}_b^f} e_c \right) - \mathcal{L}_{\mathbb{Y}_b^f} \left(\mathcal{L}_{\mathbb{X}_a} e_c \right) \\ &= \mathcal{L}_{\mathbb{X}_a} \left(G^f(\mathbb{Y}_b^f, \mathbb{Y}_c^f) \right) - \mathcal{L}_{\mathbb{Y}_b^f} \left(\mathcal{L}_{\mathbb{X}_a} e_c \right) \quad (33)\end{aligned}$$

Taking into account:

$$(\mathcal{L}_{\mathbb{X}_a} e_c)(\rho) = \frac{d}{dt} \left(\text{Tr}(e^{-\frac{i}{2} \mathbf{a} t} \rho e^{\frac{i}{2} \mathbf{a} t}) \right)_{t=0} = e_{[[c, \mathbf{a}]]}(\rho) \quad (34)$$

and the unitary invariance of G^f , we get:

$$\begin{aligned}\mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c &= G^f \left([\mathbb{X}_a, \mathbb{Y}_b^f], \mathbb{Y}_c^f \right) + G^f \left(\mathbb{Y}_b^f, [\mathbb{X}_a, \mathbb{Y}_c^f] \right) - \mathcal{L}_{\mathbb{Y}_b^f} (e_{[[c, \mathbf{a}]]}) \\ &= \mathcal{L}_{[\mathbb{X}_a, \mathbb{Y}_b^f]} e_c + G^f \left(\mathbb{Y}_b^f, [\mathbb{X}_a, \mathbb{Y}_c^f] \right) - G^f \left(\mathbb{Y}_b^f, \mathbb{Y}_{[[c, \mathbf{a}]]}^f \right) \quad (35)\end{aligned}$$

We get:

$$G^f \left(Y_{\mathbf{b}}^f, [X_{\mathbf{a}}, Y_{\mathbf{c}}^f] - Y_{[[\mathbf{c}, \mathbf{a}]]}^f \right) = 0 \quad (36)$$

This allows to conclude:

$$[X_{\mathbf{a}}, Y_{\mathbf{b}}^f] = Y_{[[\mathbf{b}, \mathbf{a}]]}^f \quad (37)$$

How does this algebra look like?

$$\mathcal{A} = \mathcal{V}_0 \oplus \mathcal{V}_0 \quad (38)$$

$$\mathcal{A} \ni \mathbf{a} = (\mathbf{a}_1, \mathbf{a}_2) \quad \mathbf{a}_1, \mathbf{a}_2 \in \mathcal{V}_0 \quad (39)$$

The properties in the previous slide allow to state:

$$\begin{aligned} [\mathbf{a}, \mathbf{b}] &= [(\mathbf{a}_1, \mathbf{a}_2), (\mathbf{b}_1, \mathbf{b}_2)] = ([\mathbf{a}_1, \mathbf{b}_1]_1, \mathbf{0}) + (\mathbf{0}, [\mathbf{a}_1, \mathbf{b}_2]_1) \\ &\quad + (\mathbf{0}, [\mathbf{a}_2, \mathbf{b}_1]_1) + [(\mathbf{0}, \mathbf{a}_2), (\mathbf{0}, \mathbf{b}_2)]_2 \end{aligned} \quad (40)$$

Where:

$$[\mathbf{a}_1, \mathbf{b}_1]_1 = [[\mathbf{a}_1, \mathbf{b}_1]] = \frac{1}{2i}(\mathbf{a}_1 \mathbf{b}_1 - \mathbf{b}_1 \mathbf{a}_1) \quad (41)$$

The bracket $[\ , \]_2$ must be determined.

A particular solution

In the qubit case we have:

$$G_f = \frac{1}{1-r^2} dr \otimes dr + \frac{r^2}{1+r} \left(f \left(\frac{1-r}{1+r} \right) \right)^{-1} (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (42)$$

For a traceless self-adjoint operator holds $\mathbf{a} = \sum_{j=1}^3 a_j \sigma_j$, so we have:

$$e_{\mathbf{a}}(\rho) = \text{Tr}(\rho \mathbf{a}) = \frac{1}{2} (a_1 r \sin \theta \cos \phi + a_2 r \sin \theta \sin \phi + a_3 r \cos \theta) \quad (43)$$

and the gradient vector field $Y_{\mathbf{a}}^f$ associated with $e_{\mathbf{a}}$ by means of G_f is

$$Y_{\mathbf{a}}^f = G_f^{-1}(de_{\mathbf{a}}, \bullet) = G_f^{-1}(\bullet, de_{\mathbf{a}}). \quad (44)$$

A particular solution

Setting $e_j \equiv e_{\sigma_j}$ and $Y_j^f \equiv Y_{\sigma_j}^f$, direct computations bring us to:

$$Y_1^f = (1 - r^2) \sin \theta \cos \phi \partial_r + g(r) \left(\cos \theta \cos \phi \partial_\theta - \frac{\sin \phi}{\sin \theta} \partial_\phi \right) \quad (45)$$

$$Y_2^f = (1 - r^2) \sin \theta \sin \phi \partial_r + g(r) \left(\cos \theta \sin \phi \partial_\theta + \frac{\cos \phi}{\sin \theta} \partial_\phi \right) \quad (46)$$

$$Y_3^f = (1 - r^2) \cos \theta \partial_r - g(r) \sin \theta \partial_\theta, \quad (47)$$

with:

$$g(r) = \frac{1+r}{r} f\left(\frac{1-r}{1+r}\right). \quad (48)$$

A particular solution

The fundamental vector fields of the action $\tilde{\alpha}$ read:

$$X_1 = -\sin \phi \partial_\theta - \cot \theta \cos \phi \partial_\phi \quad (49)$$

$$X_2 = \cos \phi \partial_\theta - \cot \theta \sin \phi \partial_\phi \quad (50)$$

$$X_3 = \partial_\phi \quad (51)$$

Where $X_j = X_{\sigma_j}$.

The commutators between the gradient vector fields can be computed:

$$[Y_1^f, Y_2^f] = F(r) X_3 \quad (52)$$

$$[Y_2^f, Y_3^f] = F(r) X_1 \quad (53)$$

$$[Y_3^f, Y_1^f] = F(r) X_2 \quad (54)$$

where we have set

$$F(r) = (1 - r^2)g'(r) + g^2(r) \quad (55)$$

A particular solution

Now, to obtain a Lie algebra, we must set $F(r) = A$ with A a constant, obtaining:

$$(1 - r^2)g'(r) + g^2(r) = A \quad (56)$$

Operating the change of variable:

$$t = \frac{1 - r}{1 + r} \quad (57)$$

And defining:

$$h(t) = \frac{f(t)}{1 - t} \quad (58)$$

We get:

$$t h'(t) - h^2(t) = -\frac{A}{4} \quad (59)$$

This is a Riccati ODE.

We will deal with the cases $A > 0$, $A = 0$ and $A < 0$ separately.

Case $A > 0$

In this case the only solution of this Riccati equation satisfying:

$$t f(t^{-1}) = f(t) \quad (60)$$

$$\lim_{t \rightarrow 1} f(t) = 1 \quad (61)$$

Is the following:

$$f_A(t) = \frac{\sqrt{A}}{2}(1-t) \frac{1+t^{\sqrt{A}}}{1-t^{\sqrt{A}}} \quad (62)$$

Case $A > 0$

With this result the monotone metric takes the following form:

$$G_A = \frac{1}{1-r^2} dr \otimes dr + \frac{r}{\sqrt{A}} \frac{(1+r)^{\sqrt{A}} - (1-r)^{\sqrt{A}}}{(1+r)^{\sqrt{A}} + (1-r)^{\sqrt{A}}} (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi) \quad (63)$$

By other means, we have that $\mathbb{X}_{\mathbf{b}} + \mathbb{Y}_{\mathbf{a}}^{f_A}$ is a fundamental vector field of an action of $\mathrm{SL}(n, \mathbb{C})$ given by:

$$\alpha_{A,g}(\rho) = \frac{(g\rho^{\frac{1}{A}}g^\dagger)^A}{\mathrm{Tr} \left((g\rho^{\frac{1}{A}}g^\dagger)^A \right)} \quad (64)$$

Note that the case $A = 1$ gives rise exactly to the Bures-Helstrom case.

Case $A = 0$

In this case, imposing the properties of operator monotone functions:

$$t f(t^{-1}) = f(t) \quad (65)$$

$$\lim_{t \rightarrow 1} f(t) = 1 \quad (66)$$

Gives rise to the following solution:

$$f_0(t) = \frac{t - 1}{\log t} \quad (67)$$

Case $A = 0$

This is the operator monotone function giving rise to B-K-M metric:

$$G_{BKM} = \frac{1}{1-r^2} dr \otimes dr + r \log \left(\frac{1+r}{1-r} \right) (d\theta \otimes d\theta + \sin^2 \theta d\phi \otimes d\phi). \quad (68)$$

The BKM case has been already solved to give rise to the following action:

$$\alpha_0(U, \mathbf{a}, \rho) = \frac{e^{U \log \rho U^\dagger + \mathbf{a}}}{\text{Tr} (e^{U \log \rho U^\dagger + \mathbf{a}})} \quad (69)$$

Notice here that in this case we don't get an action of $\text{SL}(n, \mathbb{C})$.
It is an action of $\text{SU}(n) \rtimes_R \mathcal{V}$ with $R_U(\mathbf{h}) = U\mathbf{h}U^\dagger$.

Case $A < 0$

In this case, the general solution is the following:

$$h(t) = -\frac{\sqrt{-A}}{2} \tan \left(\frac{\sqrt{-A}}{2} \log t - \frac{\sqrt{-A}}{2} c \right) \quad (70)$$

Where c is an integration constant.

So that:

$$f(t) = \frac{\sqrt{-A}}{2} (t-1) \tan \left(\frac{\sqrt{-A}}{2} \log t - \frac{\sqrt{-A}}{2} c \right) \quad (71)$$

Considerations on the regularity of this functions led to exclude this case.

The tensor G^f wouldn't be defined everywhere on S_2 .

Conclusions and outlooks

- Given a monotone metric G^f on $\mathcal{S}_n$, the vector fields $\mathbb{X}_a$ and $\mathbb{Y}_b^f$ allow in some cases to recover a group action on the space of quantum states.
- A complete description of this instance has been given in the case of the qubit.
- SPOILER: We also investigated the case $n > 2$ and we found all the group action found for the qubit ($n = 2$). However, we are not able to prove if these exhaust all possibilities when $n > 2$.

Thank You for Your attention!